SHUBHASHIS SINHA

Staff AI Architect — AI CoE | Generative AI Platforms · Agentic AI · RAG · LLM Orchestration · Cloud-Native AI

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SUMMARY

AI platform architect with **12+ years of software engineering experience**, including **4+ years designing and deploying production-grade Generative AI systems**. Currently leading AI architecture in the AI Center of Excellence at GE HealthCare — building LLM integrations, RAG pipelines, Agentic AI workflows, and cloud-native AI platforms serving enterprise-scale workloads. Deep expertise in **Python, LangGraph/LangChain, MCP protocol, Kubernetes, CI/CD**, and end-to-end AI application lifecycle from prompt engineering through evaluation, governance, and production monitoring. Proven track record of driving engineering excellence across cross-functional teams with measurable impact — $2M+ cost savings, ~40% productivity gains, and AI systems processing 1,000+ tasks/month.

CORE COMPETENCIES

Generative AI / LLMs	RAG Pipelines · Agentic AI (LangGraph, MCP, A2A, Tool-Calling) · Prompt Engineering · LLM Evaluation Frameworks · AI Governance · Model Monitoring · Responsible AI · Hallucination Mitigation · Bedrock / Claude / GPT
AI Platform Engineering	Cloud-Native AI Apps · Kubernetes · Docker · CI/CD for AI · FastAPI · Microservices · API Design · AI-Assisted Development (GitHub Copilot, Claude Code) · MLOps (MLflow, SageMaker)
Software Architecture	Distributed Systems · Event-Driven Architecture · Design Patterns · SDLC · Scalability · Reliability · Observability · Cost Optimization · Security (IAM, KMS)
Data & ML Engineering	PySpark/Scala · Kafka/MSK · AWS Glue · EMR · Redshift · Databricks · Snowflake · Apache Iceberg · Real-Time & Batch Pipelines · Vector DBs (pgvector, FAISS)
Cloud Platforms	AWS (S3, Glue, Redshift, EMR, Lambda, Step Functions, SageMaker, Bedrock, EKS, Neptune, Kinesis) · Terraform · Infrastructure as Code
Programming	Python (Expert) · .NET (Working knowledge) · SQL (Advanced) · SPARQL · Scala · Git · CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Staff AI Architect — AI Center of Excellence

Dec 2021 – Present

GE HealthCare · Bangalore, India

Generative AI Platforms & Agentic AI Systems

- Designed and deployed **end-to-end Generative AI applications** — LLM integrations, RAG pipelines, Agentic AI workflows, and AI APIs serving enterprise-scale workloads on AWS Bedrock (Claude) with **sub-second P95 latency** and production SLAs.
- Architected **multi-agent Agentic AI platform** using LangGraph with MCP protocol — autonomous agents with tool-calling, self-correction, template routing, and multi-step reasoning. Achieved **~95% task success rate** on complex orchestration workflows.
- Shipped **RAG system with hybrid retrieval** (embeddings + semantic search + keyword) across **10,000+ documents** — implemented chunking strategies, re-ranking, evidence-based reasoning, and hallucination mitigation. 20% faster retrieval vs. baseline.
- Deployed **Contract Digitalization via Agentic AI** — autonomous agents processing **1,000+ contracts/month**. Full lifecycle: requirements → architecture → prompt engineering → evaluation → production monitoring.
- Built **GenAI workflow automation** (RAG + LLM orchestration) — requirement → story → task decomposition → auto-assignment. **~40% effort reduction**, adopted by 4+ teams / 20+ developers.

AI Platform Engineering & Cloud-Native Architecture

- Designed **scalable AI platform architecture** — compute (EKS/Kubernetes), storage (S3, Iceberg), orchestration (Step Functions), serving (FastAPI, Lambda) with auto-scaling, CI/CD pipelines, containerization, and cost controls.
- Implemented **AI governance and evaluation frameworks** — prompt versioning, model evaluation pipelines (LangSmith), A/B testing, responsible AI guardrails (Bedrock Guardrails: topic, PII, content filtering), and continuous model monitoring.
- Built **reusable AI components library** — agent templates, prompt libraries, evaluation harnesses, and deployment patterns adopted across multiple product teams. **25% faster development cycles.**
- Operated **data platform processing 10TB+ daily** — 99.9% uptime across ingestion, transformation, and serving layers powering AI/ML workloads.

Engineering Excellence & Technical Leadership

- Led **architecture standards and design reviews** across the AI CoE — defined coding standards, testing strategies (unit/integration/evaluation), deployment procedures, and observability patterns for AI applications.
- Mentored **8+ engineers** on Generative AI patterns (RAG, agents, prompt engineering), cloud-native development, and software architecture best practices.
- Drove **Teradata-to-Redshift migration** — re-engineered 500+ pipelines delivering **40% cost reduction ($2M+ annual savings)** with 3x query performance improvement.

- Evaluated and integrated **emerging AI technologies** — GraphRAG (Neptune), MCP protocol, Strands SDK, AI-assisted development tools — continuously improving developer productivity and solution quality.

Tech Stack: Python, FastAPI, .NET · LangGraph/LangChain, MCP, Strands SDK · RAG/Embeddings · Vector DB (pgvector, FAISS) · AWS (Bedrock, SageMaker, EKS, Lambda, Step Functions, S3, Glue, Redshift, EMR, Neptune) · Kubernetes · Docker · Terraform · CI/CD · Kafka/MSK · Databricks · Apache Iceberg · LangSmith · MLflow · Git

Senior Software Engineer — Data & AI

Oct 2017 – Dec 2021

Bank of America · Chennai, India

- Architected **distributed batch and streaming applications** for regulatory reporting — processing **5TB+ daily** with 20% faster transformations on EMR and Snowflake.
- Designed **scalable data pipelines and ML feature stores** with incremental loading, full lineage, and data quality gates — improved accuracy by 15%, reduced platform costs by 25%.
- Built **cloud-native ETL platform** (AWS EMR + S3 + Glue + Kubernetes) — containerized workloads, CI/CD automation, improved reliability and developer productivity.
- Optimized **complex SQL and Spark transformations** across 100B+ row datasets — 40% reduction in execution time through partition optimization and predicate pushdown.
- Collaborated with **cross-functional teams** (risk, compliance, trading) to define solution architectures, translate business requirements into technical implementations, and deliver reliable data products.

Tech Stack: Python, Scala, SQL · Apache Spark · Snowflake · Kafka · AWS (S3, EMR, Glue) · Kubernetes · Docker · Hive · Hadoop · CI/CD

Software Engineer — Data Platform

Mar 2014 – Oct 2017

T-Mobile (via HCL Technologies) · Bangalore, India

- Built **data pipelines at scale** (Hadoop, Hive, Spark) across **5M+ subscribers** — enterprise integration workflows connecting CRM, billing, and network systems.
- Implemented **real-time ingestion** (CDC/streaming) — reduced data freshness from T+1 to near real-time for downstream analytics and ML feature pipelines.
- Developed **Python automation tools** for data validation, monitoring, and alerting — 25% improvement in operational efficiency.

Tech Stack: Python, SQL, Hadoop, Hive, Sqoop, Spark, Shell Scripting

EDUCATION

Bachelor of Technology — Electronics & Communications Engineering

West Bengal University of Technology · Kolkata, India · GPA: 8.87/10

AWARDS & CERTIFICATIONS

Sir M. Visvesvaraya Award — GE HealthCare (2024)

Recognized for building production Generative AI platforms with measurable enterprise impact across the AI CoE.

Impact Award — GE HealthCare (Oct 2022, Nov 2023, Mar 2024)

Platform migration ($2M+ savings), reusable AI frameworks, agentic automation adopted by multiple teams.

AWS Certified Solutions Architect – Associate (2025)